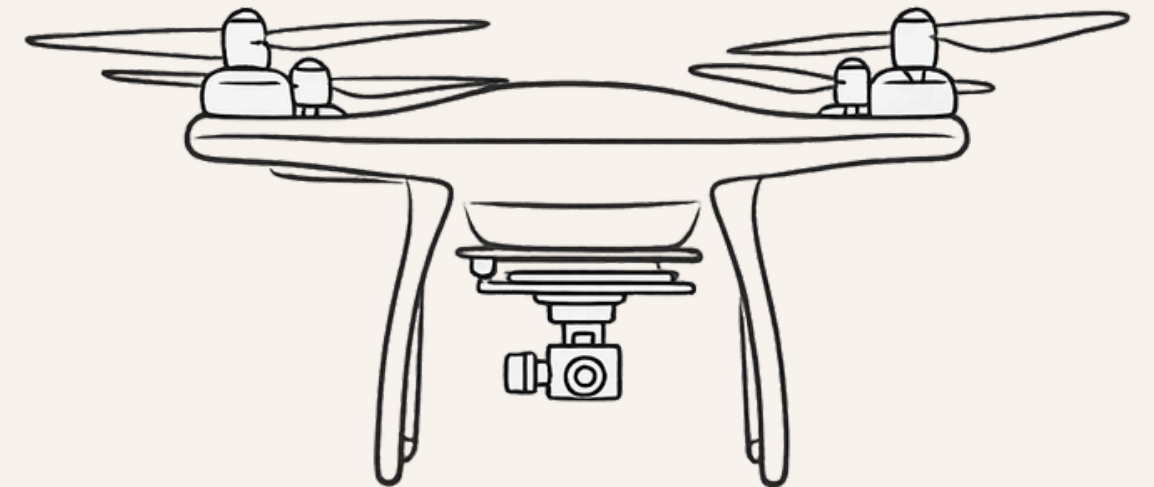
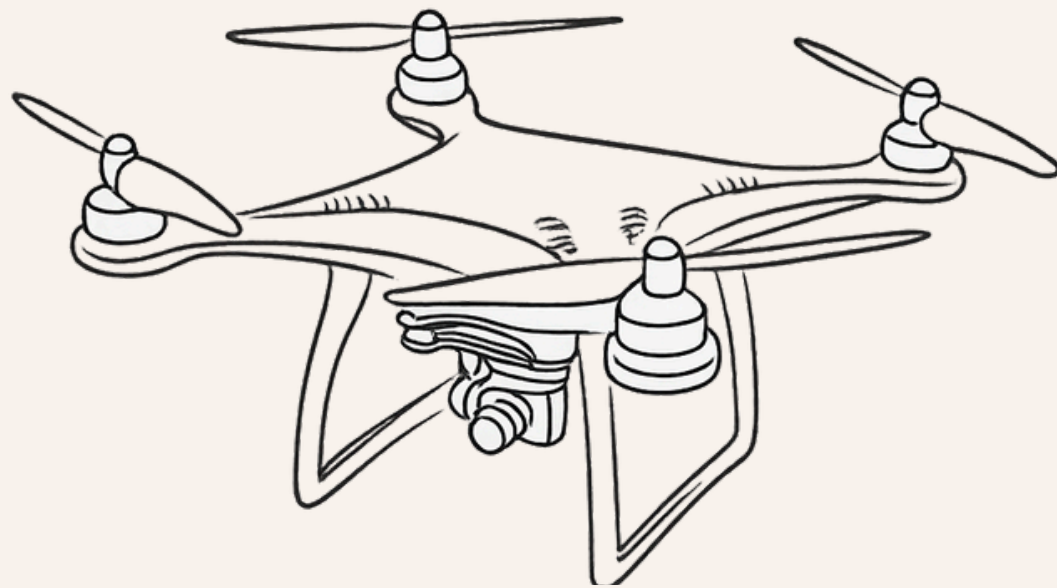


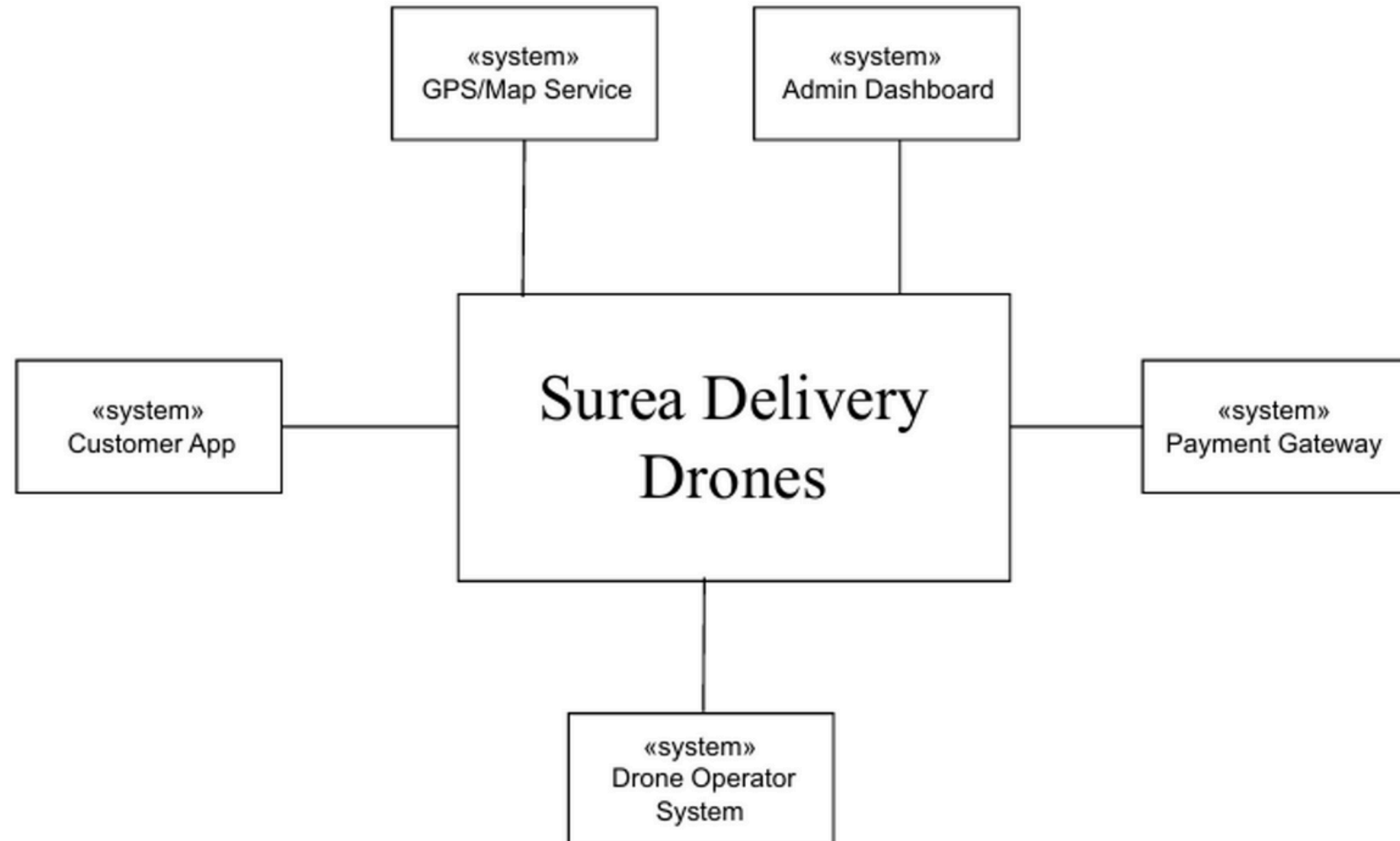
SUREA DELIVERY DRONES



Surea Delivery Drones

is a system that manages a fleet of drones to deliver parcels locally. It automates order scheduling and drone assignment, optimizes routes for safety and battery limits, and monitors deliveries in real time. Customers receive delivery status updates and proof-of-delivery via OTP or QR codes. The system also provides dashboards for admins and maintenance notifications for drone technicians, using simulated telemetry and mock external services for maps, weather, and notifications.

Context Diagram:



System requirement:

User Functional Requirements:

- 1.The system shall allow a customer to register and create a delivery order with pickup and delivery addresses and parcel details.
- 2.The system shall provide an estimated price for the delivery based on distance and parcel weight.
- 3.The system shall allow the customer to track their parcel in real time and view live drone movement on a map.
- 4.The system shall send notifications to the customer when the delivery status changes to In Progress, Delivered, or Failed.
5. The system shall provide a user-friendly FAQ section to assist customers with common inquiries.
6. The system shall enable customers to rate their delivery experience and provide feedback.
7. The system shall allow administrators to view all active drones, their battery levels, and assigned deliveries on a centralized dashboard.
8. The system shall enable drone technicians to receive automatic maintenance alerts when a drone's battery health or flight stability drops below safe thresholds.
9. The system shall allow customers to choose between standard and express drone delivery options
- 10.The system shall automatically notify customers when weather conditions delay delivery

System requirement:

System Functional Requirements (Multilevel)

1. Customer Order Creation

- 1.1 The system shall allow input of pickup and delivery addresses, parcel type, and customer details.
- 1.2 The system shall validate addresses and required fields before confirming the order.

2. Real-Time Tracking

- 2.1 The system shall provide a live map interface showing the current location of the drone.
- 2.2 The system shall update the drone's location on the map every 30 seconds.

3. Notifications

- 3.1 The system shall allow customers to opt-in for notifications via SMS, email, or app alerts.
- 3.2 The system shall notify customers of any status changes (e.g., In Progress, Delivered, Failed).

4. Drone Monitoring

- 4.1 The system shall display all active drones on an admin dashboard with real-time telemetry data such as altitude, battery percentage, and signal strength.
- 4.2 The system shall highlight drones with low battery or signal loss by changing their status indicator color to red.

5. Weather Delay Notification

- 5.1 The system shall integrate with a live weather API to detect severe or unsafe flying conditions along the drone's route.
- 5.2 The system shall automatically send real-time notifications to customers and admins via SMS and email if a delivery is delayed, rescheduled, or canceled due to weather conditions.

System requirement: Non-Functional Requirements (NFRs)

1. Performance

- Goal: The system shall update shipment tracking locations in near real time.
- Metric: 95% of tracking API responses shall occur within $\leq$ 2 seconds.

2. Reliability

- Goal: The system shall operate continuously during normal use without failure.
- Metric: The system shall maintain at least 95 % uptime during 50 concurrent simulated deliveries.

3. Security

- Goal: The system shall protect user data and transaction information.
- Metric: The system shall implement encryption for all sensitive data in transit and at rest, achieving 100% compliance with industry standards.

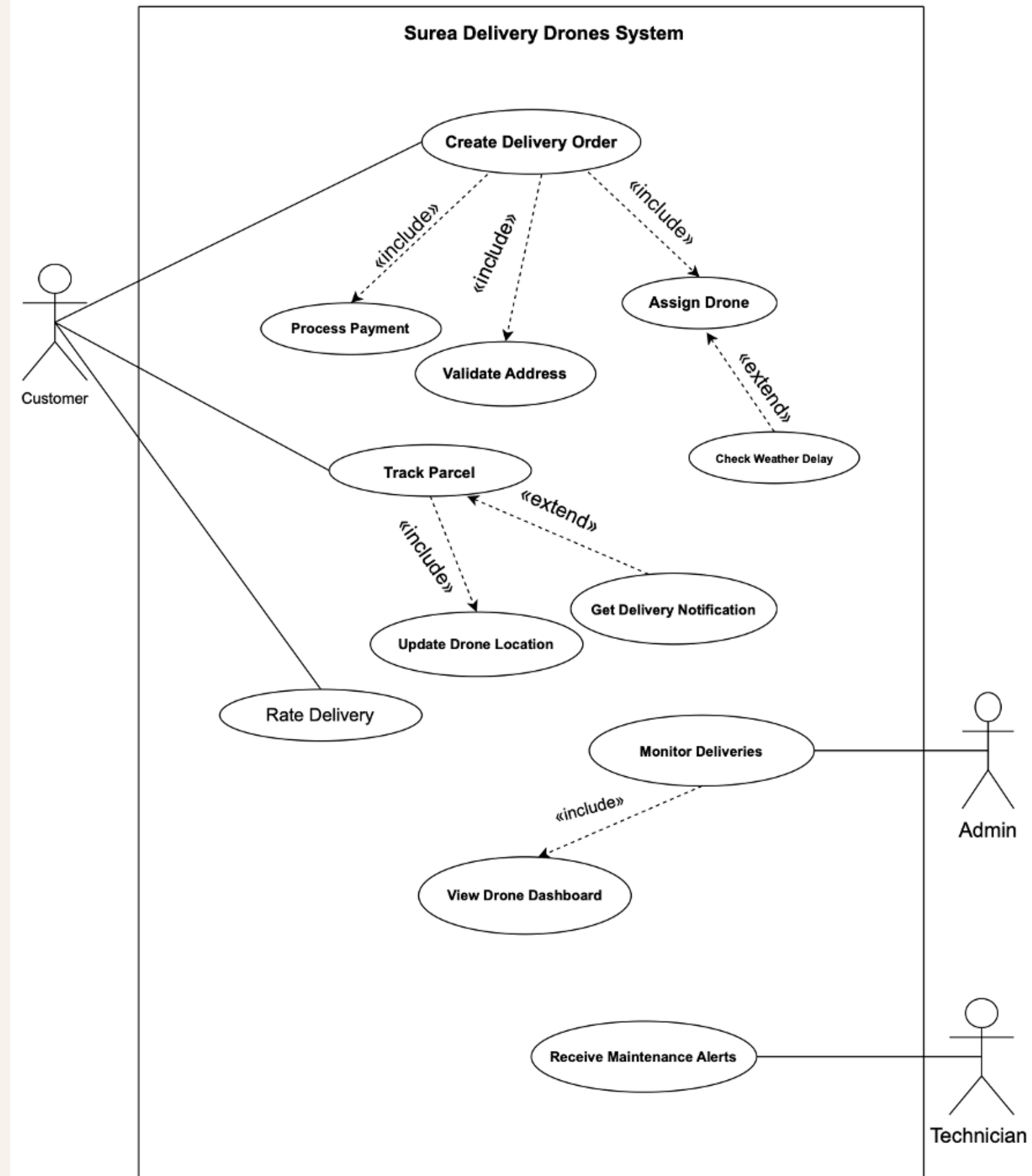
4. Usability

- Goal: The admin and technician dashboards shall be easy to use and navigate.
- Metric: At least 90% of users shall successfully complete basic tasks (e.g., viewing drone status or acknowledging alerts) during usability testing.

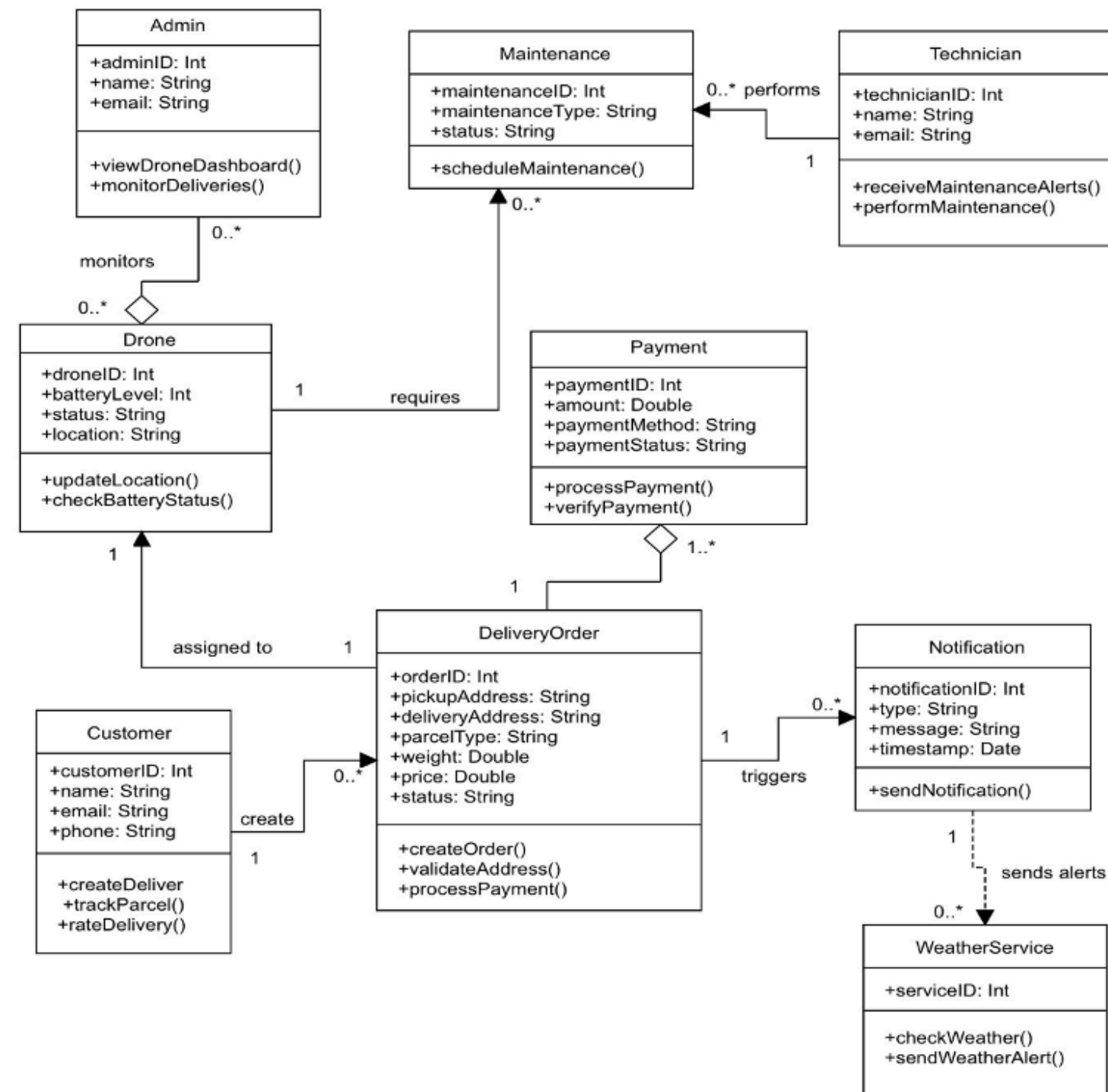
5. Scalability

The system shall support growth in both users and drones without any performance degradation, and shall handle up to 10 000 concurrent user requests while maintaining an average response time of 3 seconds or less.

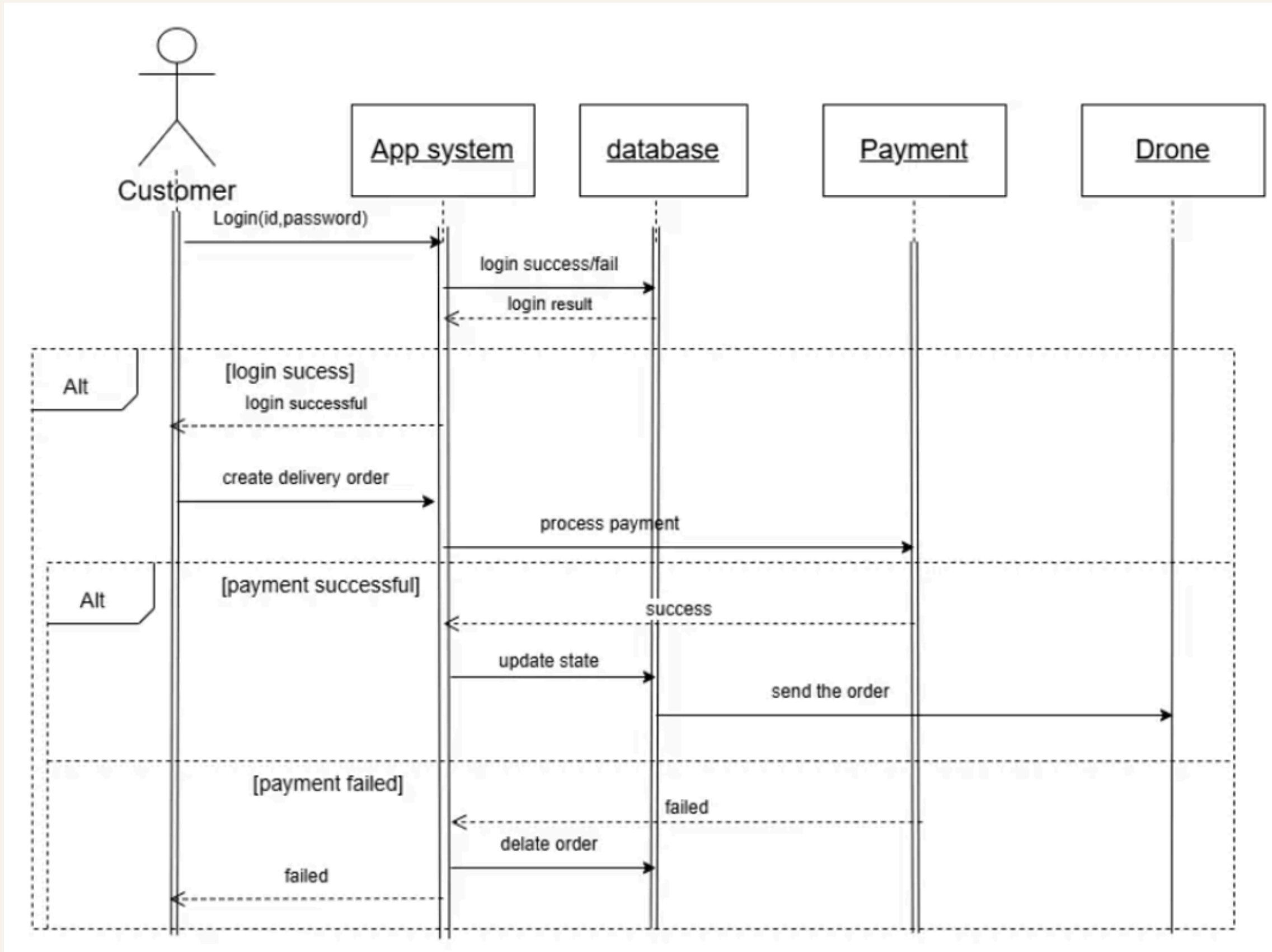
Use-case Diagram



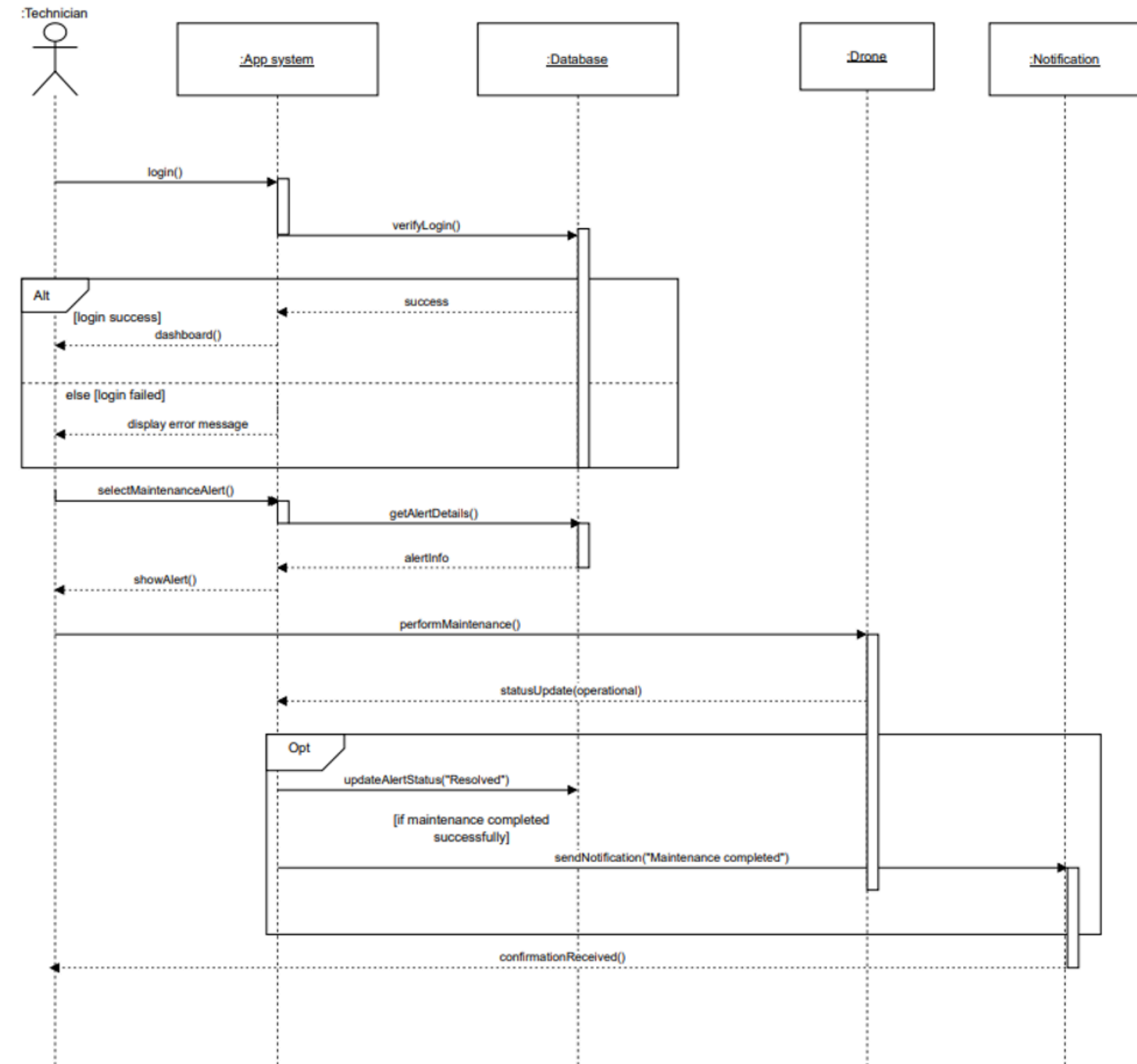
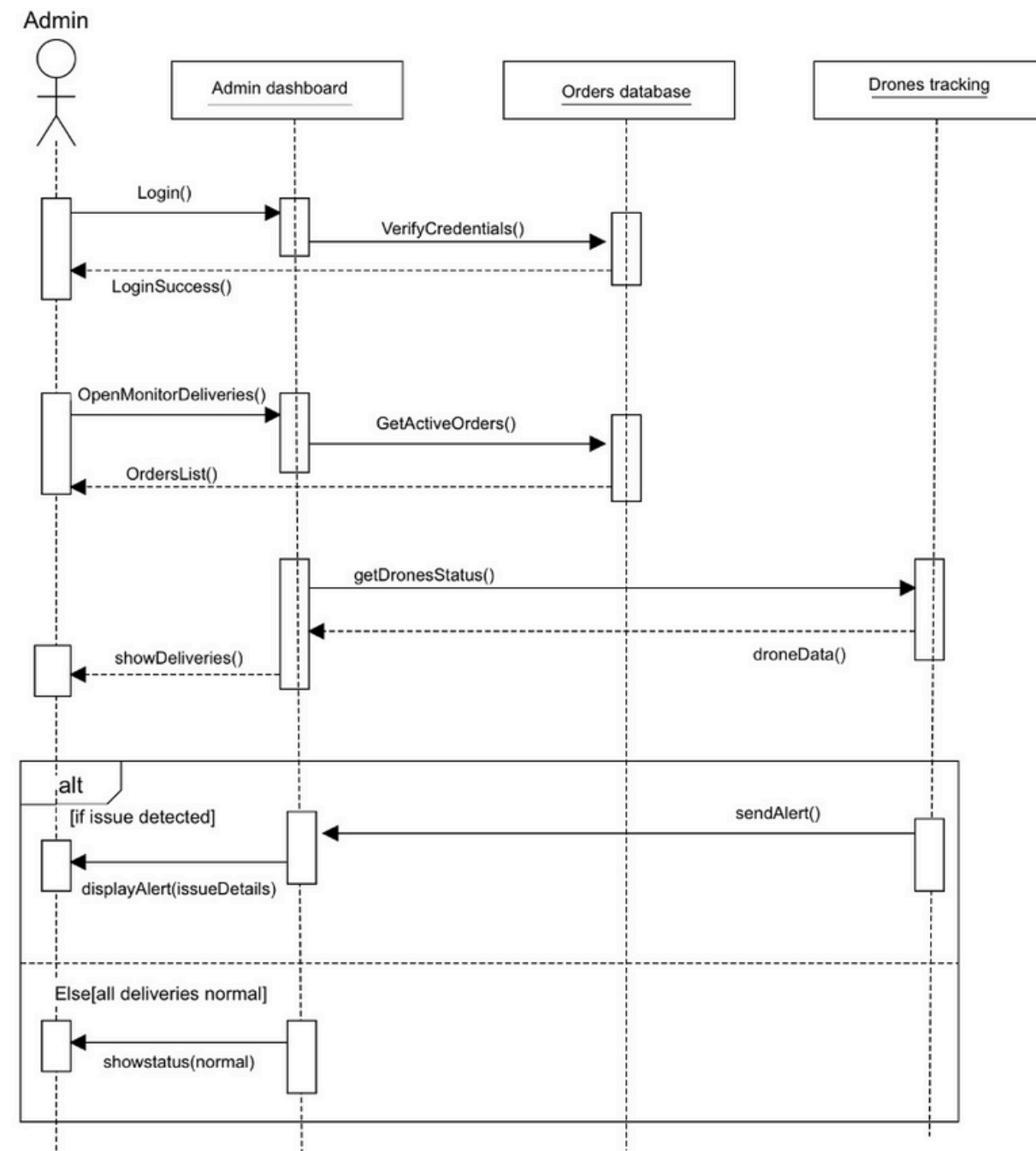
System Class Diagram



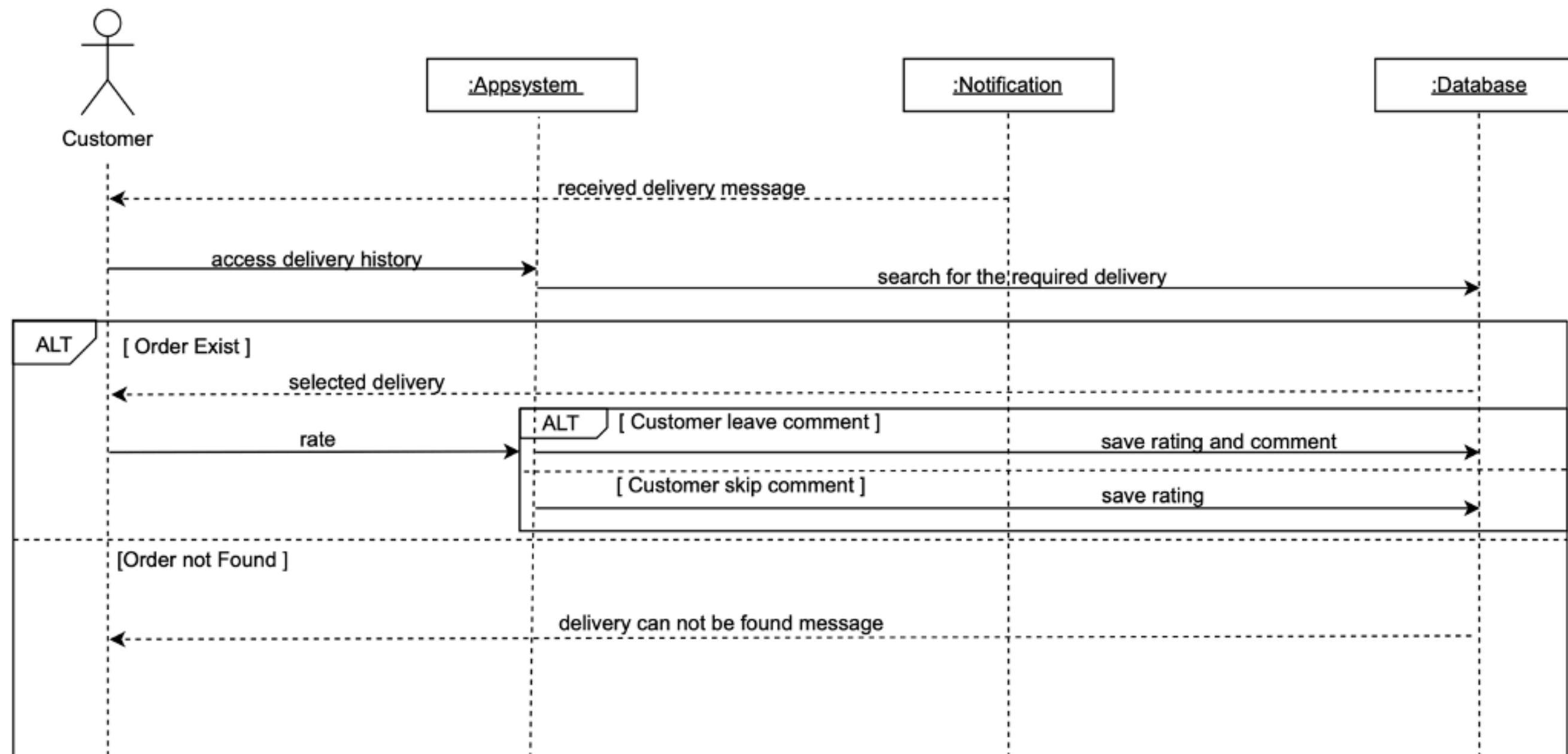
Sequence Diagram



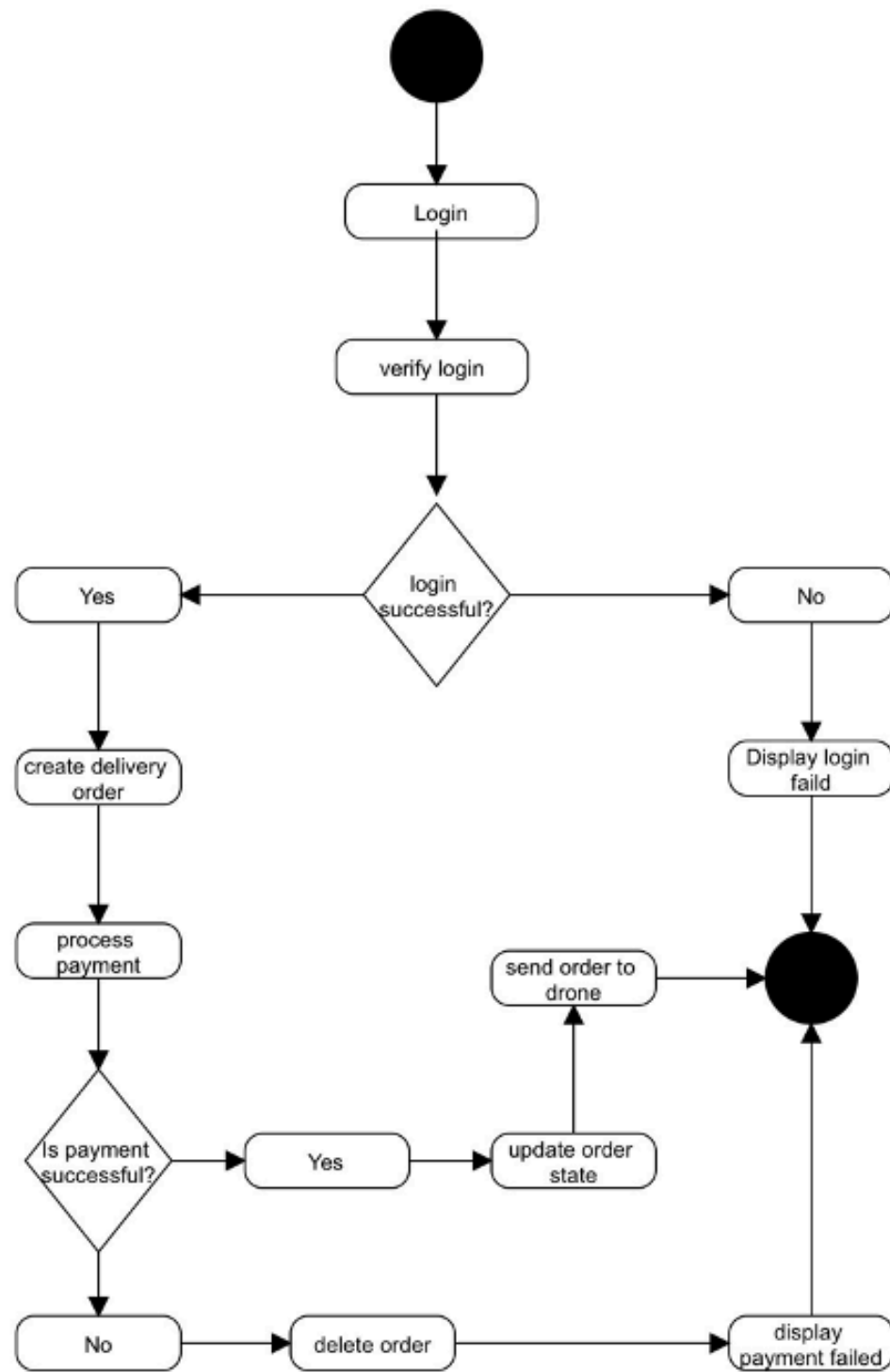
Sequence Diagram



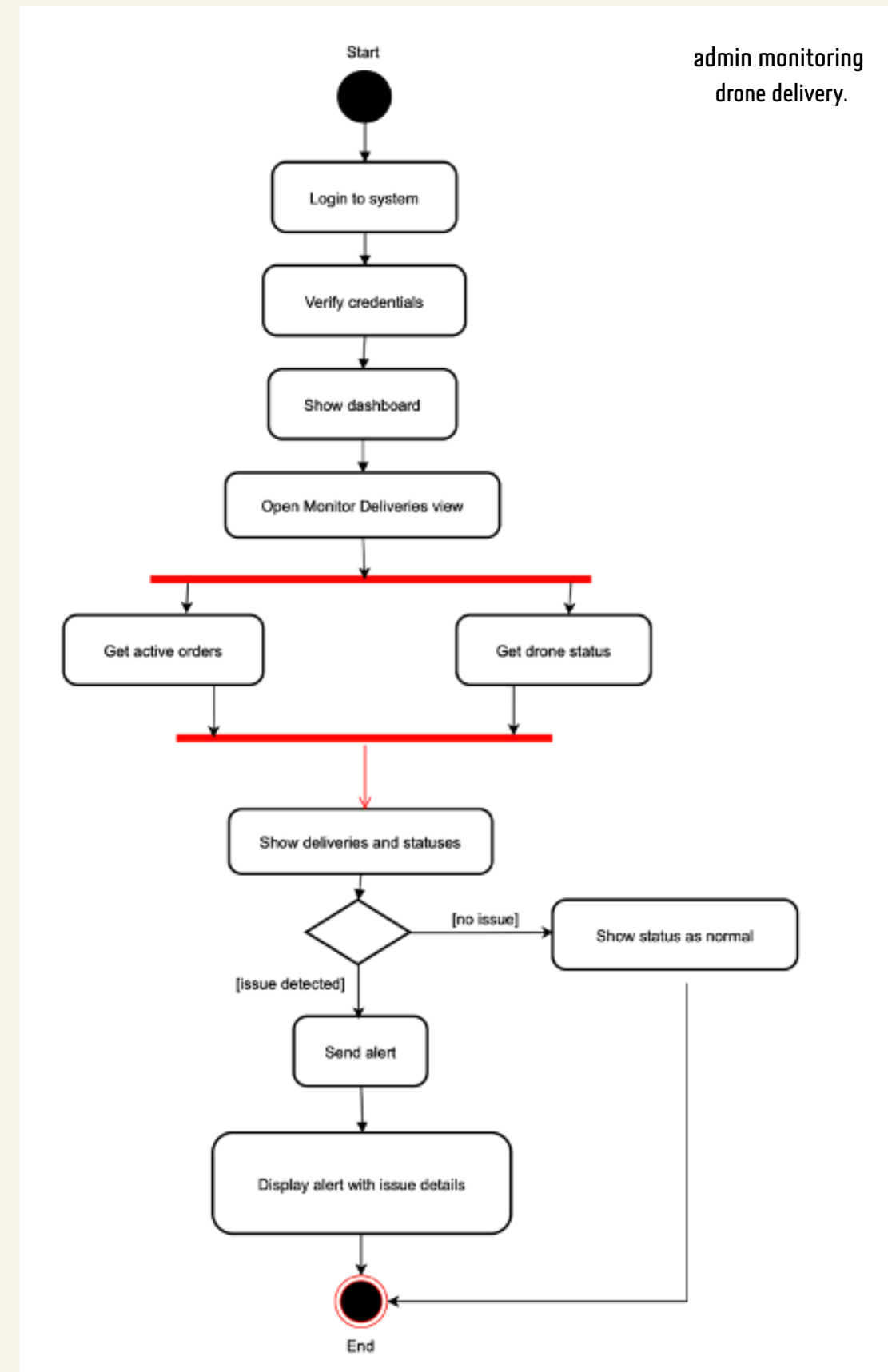
Sequence Diagram



Activity Diagram



Activity Diagram



Architecture Diagram

